

Exercise 2.1: Question 1

Comprehensive step-by-step solutions for all 6 subdivisions of the first simplification
sum

SUBDIVISIONS 1 & 2

SUBDIVISION 1 Simplify $i^{1947} + i^{1950}$

Apply division algorithm to reduce powers modulo 4:

- $1947 = 4 (486) + 3 \Rightarrow i^{1947} = i^3$

- $1950 = 4 (487) + 2$

$\Rightarrow$

$$i^{1950}$$

$=$

$$i^2$$

$$\begin{aligned} i^{1947} + i^{1950} &= i^3 + i^2 \\ &= -i - 1 = -1 - i \end{aligned}$$

SUBDIVISION 2 Simplify $i^{1948} - i^{-1869}$

Analyze positive and negative exponents modulo 4:

- $1948 = 4 (487) + 0 \Rightarrow i^{1948} = i^0 = 1$

- $-1869 = 4 (-468) + 3 \Rightarrow i^{-1869} = i^3 = -i$

$$\begin{aligned} i^{1948} - i^{-1869} &= 1 - (-i) \\ &= 1 + i \end{aligned}$$

SUBDIVISIONS 3 & 4

SUBDIVISION 3 Simplify $\sum_{n=1}^{12} i^n$

We use the cyclic property that the sum of any four consecutive powers of i is zero:

$$i^k + i^{k+1} + i^{k+2} + i^{k+3} = 0$$

The limit 12 contains exactly three full blocks of 4 terms:

$$\begin{aligned} \sum_{n=1}^{12} i^n &= (i + i^2 + i^3 + i^4) \\ &+ (i^5 + \dots + i^8) + (i^9 + \dots + i^{12}) \\ &= 0 + 0 + 0 = 0 \end{aligned}$$

SUBDIVISION 4 Simplify $i^{59} + \frac{1}{i^{59}}$

First, reduce the positive index 59 modulo 4:

- $59 = 4(14) + 3 \Rightarrow i^{59} = i^3 = -i$
- The reciprocal is $\frac{1}{-i} = \frac{i}{-i^2} = i1 = i$

$$\begin{aligned} i^{59} + \frac{1}{i^{59}} &= (-i) + (i) \\ &= 0 \end{aligned}$$

SUBDIVISION 5

PRODUCT Exponent Addition Rule

For matching bases, we sum up the exponents from 1 to 2000:

$$i \cdot i^2 \cdot i^3 \cdots i^{2000} = i^{1+2+3+\cdots+2000}$$

We apply the sum of first n natural numbers formula:

$$\text{Sum} = \frac{n(n+1)}{2}$$

SOLUTION Exponent Calculation

Calculate the total sum and reduce modulo 4:

$$\begin{aligned}\text{Sum} &= \frac{2000 \cdot 2001}{2} = 1000 \cdot 2001 \\ &= 2001000\end{aligned}$$

$$\begin{aligned}i^{2001000} &= i^{4(500250)} \\ &= (i^4)^{500250} = 1^{500250} = 1\end{aligned}$$

SUBDIVISION 6

FACTORING Simplify $\sum_{n=1}^{10} i^{n+50}$

Pull out the constant base factor i^{50} from the sum:

$$\sum_{n=1}^{10} i^{n+50} = i^{50} \sum_{n=1}^{10} i^n$$

Evaluate both components separately:

$$\bullet i^{50} = i^{4(12) + 2} = i^2 = -1$$

EXPANSION Series Evaluation

Now evaluate the remaining sum of 10 terms:

$$\begin{aligned} \sum_{n=1}^{10} i^n &= (i + i^2 + i^3 + i^4) \\ &+ (i^5 + \dots + i^8) + i^9 + i^{10} \\ &= 0 + 0 + i - 1 = -1 + i \end{aligned}$$

$$\text{Total} = -1 \cdot (-1 + i) = 1 - i$$